F5.2 – Environmental Drivers vs Energy Consumption

Sprint 2

Executive Summary

Purpose. F5.2 investigates how indoor temperature, humidity, luminosity and occupancy are associated with the different energy load types in the building. The analysis was carried out for NW, SE and SW; NE was excluded from the indoor environmental analysis because the environmental sensor values are missing.

Approach. For every available driver–load combination, Pearson and Spearman correlations were calculated, followed by partial correlations controlling for hour of day. Scatter plots with LOWESS smoothing were used to inspect non-linear patterns, and an occupancy-specific boxplot was added where a binary predictor made LOWESS less informative.

Main message. Environmental effects are strongly zone- and load-dependent. Several associations remain after controlling for hour, but some statistically strong results are not physically intuitive and therefore should be interpreted as associations rather than causal effects.

1. Objective and Data Preparation

- The cleaned hourly dataset was filtered to observations with usable_for_model = 1.
- The working dataset contained 437,158 rows and four zones (NE, NW, SE, SW).
- The load types available were HVAC, Lighting, Plugs & Sockets, Emergency Lighting, Hand Dryer and RACK.
- Environmental variables used were indoor temperature (temp_c), relative humidity (hum_pct), luminosity (lum_lux) and occupancy.
- Environmental measurements were reduced to one record per timestamp and zone, while energy loads were pivoted to one column per load type. The resulting load table contained 98,631 timestamp–zone rows.
- NW, SE and SW were used for F5.2. NE was excluded from indoor-environment relationships because its environmental values are missing.

2. Methodology

- **Pearson correlation:** used to quantify linear association between each environmental driver and each load type.
- **Spearman correlation:** used to capture monotonic relationships that may not be linear.
- **Partial correlation controlling for hour:** hour of day was treated as a categorical control variable. The hour effect was removed from both the driver and load before correlating the residuals.
- **LOWESS plots:** used to visually inspect non-linear shapes. For plotting only, a deterministic sample of up to 3,000 observations was used; the correlation statistics were calculated on the full available data.
- **Occupancy comparison:** because occupancy is binary, mean/median comparisons and a boxplot were used in addition to correlations where appropriate.

3. Main Relationships

The table below reports the relationships with an absolute partial correlation of at least 0.20 after controlling for hour of day.

Zone	Environmental driver	Load type	Pearson	Spearman	Partial corr. (hour)
NW	Temperature	Plugs & Sockets	-0.439	-0.393	-0.457
NW	Luminosity	Plugs & Sockets	-0.061	-0.018	-0.287
NW	Temperature	Emergency Lighting	-0.230	-0.472	-0.226
SE	Temperature	Emergency Lighting	0.794	0.835	0.798
SE	Luminosity	Lighting	0.318	0.145	0.275
SE	Occupancy	Plugs & Sockets	-0.260	-0.279	-0.273
SE	Occupancy	Emergency Lighting	0.236	0.265	0.240
SW	Occupancy	Lighting	0.317	0.206	0.297
SW	Luminosity	RACK	0.150	-0.021	0.229

4. Interpretation by Zone

NW. The clearest result is the negative association between indoor temperature and Plugs & Sockets consumption (partial correlation = -0.457). The LOWESS curve shows a strong decrease at lower temperatures and a flatter pattern above approximately 20–22°C. NW also shows the expected negative luminosity–Lighting relationship in the raw correlations, although it becomes weaker after controlling for hour, indicating that part of the effect is explained by the daily cycle.

SE. The strongest statistical association is indoor temperature versus Emergency Lighting (partial correlation = 0.798). The LOWESS curve confirms a strong positive pattern, but the relationship is not physically intuitive; it may reflect seasonality, operational changes or another confounding factor. Temperature versus HVAC shows a more plausible decreasing, non-linear pattern: Spearman is substantially more negative than Pearson, consistent with a monotonic relationship that is not well represented by a straight line.

SW. Most environmental relationships are weak. The clearest result is occupancy versus Lighting (partial correlation = 0.297). Average Lighting power is about 44 W when occupancy is not detected and about 350 W when occupancy is detected. The median remains low in both groups, showing that the distribution is highly skewed and that occupancy is associated with occasional much higher Lighting use rather than a uniform increase at every occupied hour. Humidity generally shows weak associations with the analysed load types across all available zones.

5. Correlation Heatmaps

Pearson heatmaps provide the initial linear overview. The partial-correlation heatmaps show which relationships remain after controlling for hour of day.

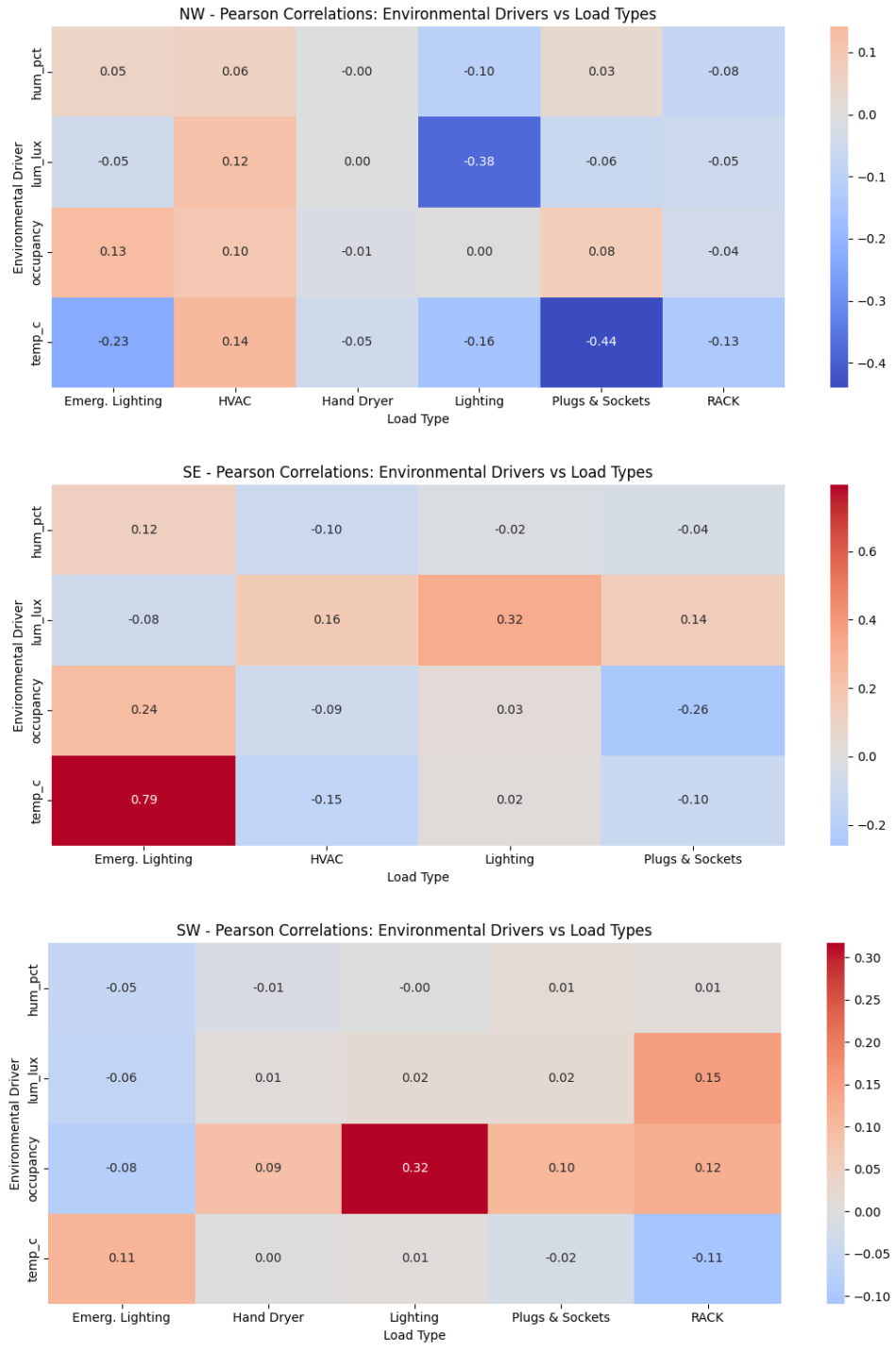


Figure 1. Pearson correlation heatmaps for NW, SE and SW.

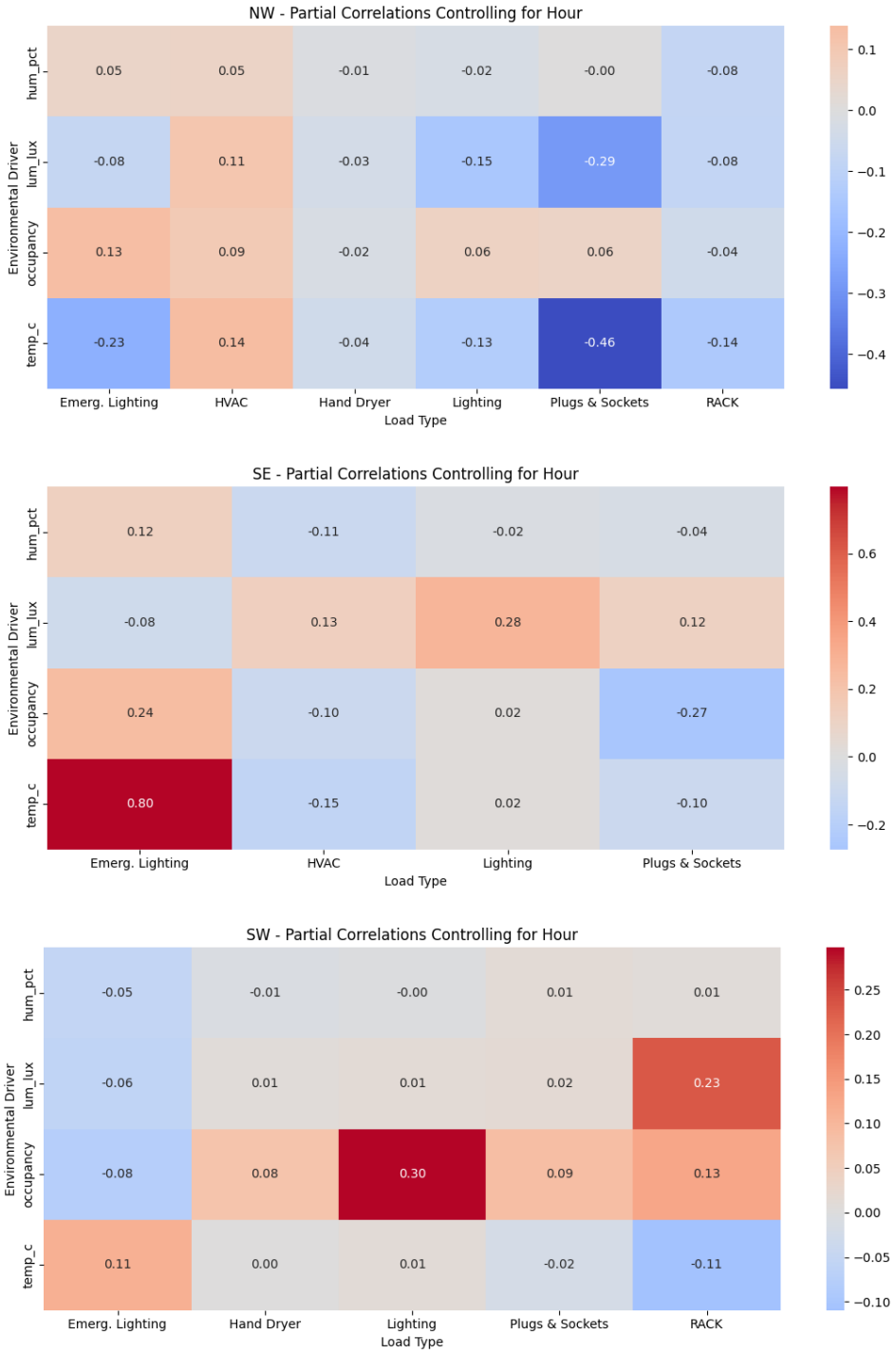


Figure 2. Partial correlations controlling for hour of day for NW, SE and SW.

6. Selected Relationship Plots

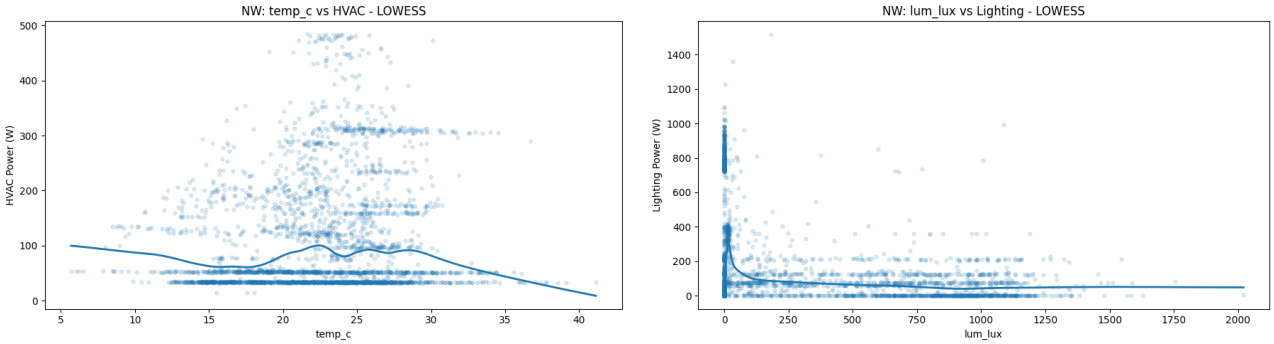


Figure 3. NW: temperature vs HVAC (left) and luminosity vs Lighting (right). The first relationship is non-linear; the second shows decreasing Lighting use as natural light increases.

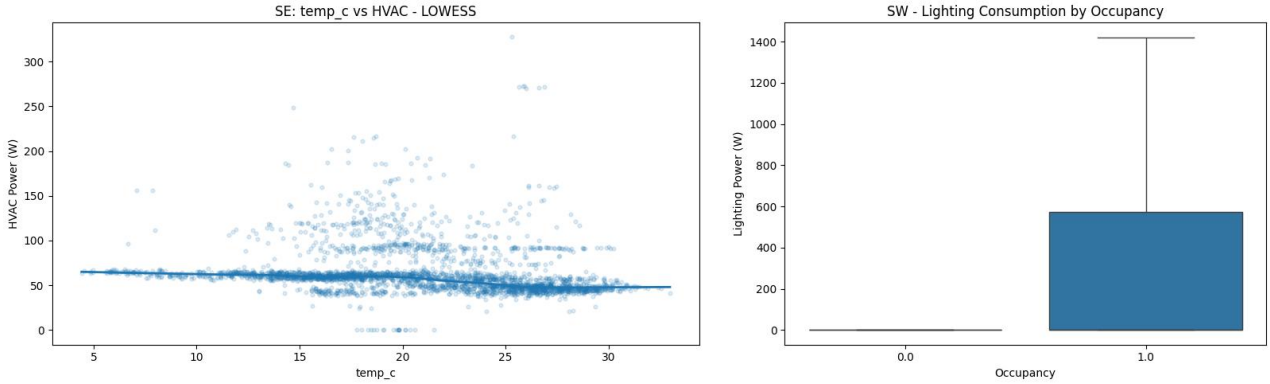


Figure 4. SE temperature vs HVAC (left) and SW occupancy vs Lighting (right). SE shows a decreasing HVAC trend, while SW Lighting is substantially higher when occupancy is detected.

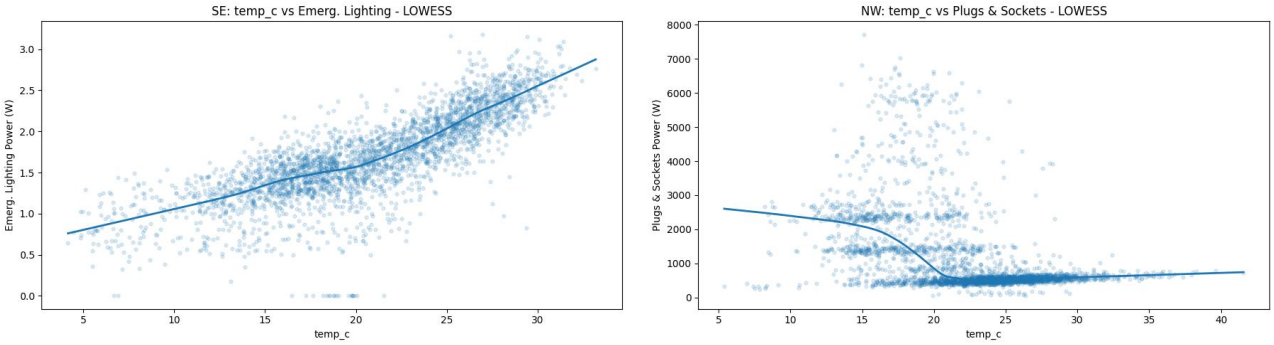


Figure 5. Two relationships requiring cautious interpretation: SE temperature vs Emergency Lighting (left) and NW temperature vs Plugs & Sockets (right). Both are statistically strong, but neither should be treated as causal without further controls.

7. Conclusion

The F5.2 analysis was performed on NW, SE and SW only. NE was excluded because the cleaned dataset does not contain valid indoor environmental sensor measurements for temperature, humidity, luminosity and occupancy in this zone. Therefore, environmental driver–load relationships could not be estimated reliably for NE.F5.2 shows that indoor environmental variables do not affect energy consumption uniformly across the building. The relevant drivers depend on both zone and load type, and several relationships are non-linear or partially explained by temporal structure. Controlling for hour changes the interpretation of some raw correlations, while other associations remain strong. These findings support the use of zone-specific environmental and temporal features in subsequent forecasting models.The analysis also highlights an important modelling caution: a strong correlation is not necessarily causal. Relationships such as SE temperature versus Emergency Lighting should be retained as diagnostic findings, but should not be interpreted as a direct physical effect without testing additional seasonal and operational controls.